IEEE CS Bangalore Chapter Internship and Mentorship Program – 2026

Duration: 1st April 2026 to 30th September 2026
Report Period: 1st May 2026 – 30th June 2026

Monthly Progress Report – 2

Project Details

Field	Value
Project ID	P107
Project Title	ASL–ISL Translation Engine (AITE)
Students	Naman Nagar, Monisha Sharma, Nandita R Nadig
University	PES University
Mentor	Dr. Sudhamani M J
Mentor's Organization	PES University

Progress of the Project

1. Problem Definition

Deaf individuals rely on sign languages such as ASL and ISL for communication. However, significant differences exist between these languages in terms of grammar, syntax, and gestures. Existing systems primarily focus on individual sign languages and do not provide real-time ASL to ISL translation.

The project aims to develop a three-stage modular system consisting of:

1. Sign Recognition.
2. Cross-lingual Translation.
3. Sign Generation.

During the current reporting period, emphasis was placed on improving the recognition component and preparing datasets for robust ISL translation.

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2. Literature Review

The literature survey conducted during the previous month was extended by studying transformer-based sign recognition systems and pose-based recognition techniques.

Additional studies included:

- SPOTER architecture.
- Pose Transformer methods.
- Holistic skeleton extraction.
- Sign normalization techniques.
- Dataset augmentation strategies.

The review confirmed that current systems suffer from:

- Limited signer generalization.
- Small datasets.
- Lack of ISL support.
- Poor real-time performance.

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3. Existing System

Existing systems mainly:

- Use hand-only keypoint extraction.
- Depend on small datasets.
- Support only ASL recognition.
- Ignore signer variations.
- Lack ISL grammar translation.

Public datasets such as WLASL contain limited training samples, increasing the possibility of overfitting.

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4. Proposed System

The proposed AITE system was further enhanced during this phase.

Major improvements include:

- MediaPipe Holistic-based extraction.
- 27-keypoint skeletal representation.
- Signing-space normalization.
- Transformer-based recognition.
- FastAPI deployment architecture.
- Custom ISL dataset planning.

The complete system consists of:

1. Video Input.

2. Keypoint Extraction.
3. Recognition Module.
4. Translation Module.
5. Sign Generation Module.

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5. Knowledge Gained – Tools, Technologies and Frameworks

The team gained practical experience in:

Area	Tools and Technologies
Pose Estimation	MediaPipe Holistic
Deep Learning	PyTorch
Transformer Models	SPOTER Architecture
Computer Vision	OpenCV
APIs	FastAPI
Data Processing	NumPy
Configuration	YAML
Model Evaluation	WER, Accuracy Metrics

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6. Architectural Framework

The architecture evolved from a hand-keypoint model to a holistic pose-based system.

Recognition Pipeline

Video Input
↓
MediaPipe Holistic
↓
27 Keypoint Extraction
↓
Normalization
↓
Transformer Encoder
↓
Gloss Prediction
↓
Translation Module
↓
ISL Generation

The updated framework improves signer independence and reduces input dimensionality.

7. Project Implementation

The following developments were completed during May–June:

Recognition Model Development

- Implemented Transformer architecture.
- Added positional encoding.
- Added attention-based decoder.

Keypoint Extraction

- Upgraded from hand-only landmarks to 27-point holistic skeleton.
- Added normalization techniques.

Dataset Processing

- Multi-format dataset loader.
- Keypoint caching.
- Frame sampling.

Data Augmentation

- Random mirror.
- Rotation.
- Scaling augmentation.

Training Pipeline

- Early stopping.
- Checkpoint saving.
- Validation monitoring.
- CSV logging.

API Development

- Prediction endpoints.
- Model loading.
- Error handling.

Custom Dataset Preparation

- ISL dataset planning.
- Annotation design.
- Quality control protocols.

8. Results

The following outcomes were achieved:

Component	Status
Transformer model	Completed
27-keypoint extraction	Completed
Dataset loader	Completed
Augmentation pipeline	Completed
Evaluation framework	Completed
API integration	Completed
Custom dataset planning	In Progress

Additional observations:

- Input dimensionality reduced significantly.
- Training speed improved through caching.
- Better signer normalization achieved.
- Framework prepared for large-scale training.

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9. Conclusion and Future Work

The second phase of the project successfully completed the recognition framework.

Future activities include:

- ISL dataset collection.
- Recognition model training.
- Translation module development.
- Avatar generation.
- Frontend integration.
- Research paper preparation.

Overall project completion is estimated at approximately **40–45%**.

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10. Research Article Preparation

The following activities have been initiated:

- Methodology documentation.
- Architecture description.

- Evaluation metrics preparation.
- Experimental framework planning.

The research paper will be completed after final experimental validation.

Signatures

Mentee's Signatures

Name	Signature	Date
Naman Nagar		
Monisha Sharma		
Nandita R Nadig		

Approved By

Name	Signature	Date
Dr. Sudhamani M J		